# Install JDK

Java Archive Downloads - Java SE 21

https://download.oracle.com/java/21/archive/jdk-21.0.7_windows-x64_bin.msi (sha256)

Also make sure Python is also installed in your system

Windows installer (64-bit) — Recommended

# Install Jenkins: Jenkins.war

Open Command Prompt and check the Java version

```
C:\Users\sweta>java -version
java version "21.0.7" 2025-04-15 LTS
Java(TM) SE Runtime Environment (build 21.0.7+8-LTS-245)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.7+8-LTS-245, mixed mode, sharing)
```

Then Type: java -jar "C:\Program Files\jenkins.war"

```
C:\Users\sweta>java -jar "C:\Program Files\jenkins.war"
Running from: C:\Program Files\jenkins.war
webroot: C:\Users\sweta\.jenkins\war
2026-01-02 14:31:32.142+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-01-02 14:31:34.849+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-01-02 14:31:34.932+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm 21.0.7+8-LTS-245
2026-01-02 14:31:36.154+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-01-02 14:31:36.245+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-01-02 14:31:37.030+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\sweta\.jenkins found at: $user.home/.jenkins
2026-01-02 14:31:37.191+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@2ce45a7b{Jenkins v2.528.3,/,b=file:///C:/Users/sweta/.jenkins/war/,a=AVA
ILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@153d4abb{STARTED}}
2026-01-02 14:31:37.252+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bbbb6a8{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-01-02 14:31:37.305+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@6853425f{STARTING}[12.0.25,sto=0] @6118ms
2026-01-02 14:31:37.307+0000 [id=45]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-01-02 14:31:37.472+0000 [id=41]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.528.3
2026-01-02 14:31:37.606+0000 [id=52]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-01-02 14:31:37.633+0000 [id=72]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-01-02 14:31:39.019+0000 [id=65]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-01-02 14:31:39.029+0000 [id=58]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-01-02 14:31:39.030+0000 [id=68]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-01-02 14:31:39.358+0000 [id=52]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-01-02 14:31:39.359+0000 [id=65]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-01-02 14:31:39.360+0000 [id=71]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-01-02 14:31:39.362+0000 [id=55]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-01-02 14:31:39.400+0000 [id=87]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-01-02 14:31:39.912+0000 [id=64]    INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> *************************************************************
[CRLF]> *************************************************************
[CRLF]> *************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 2a8107bf414a4ea8b729cf86c9573781
```

```
[CRLF]>
[CRLF]> This may also be found at: C:\Users\sweta\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> *************************************************************
[CRLF]> *************************************************************
[CRLF]> *************************************************************

2026-01-02 14:31:45.394+0000 [id=64]    INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-01-02 14:31:45.429+0000 [id=41]    INFO    hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-01-02 14:31:46.966+0000 [id=87]    INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-01-02 14:31:46.967+0000 [id=87]    INFO    hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
```

Password: 2a8107bf414a4ea8b729cf86c9573781

```
This may also be found at: C:\Users\sweta\.jenkins\secrets\initialAdminPassword
```

Save this password

Open Local Host: http://localhost:8080

Follow the steps, Add User name password, email and continue. By default local host 8080 will be there

# Jenkins is ready!

Your Jenkins setup is complete.

Start using Jenkins

localhost:8080

Update

Chat

Jenkins

New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status 0/2

Add description

## Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

**Start building your software project**

Create a job

**Set up a distributed build**

Set up an agent

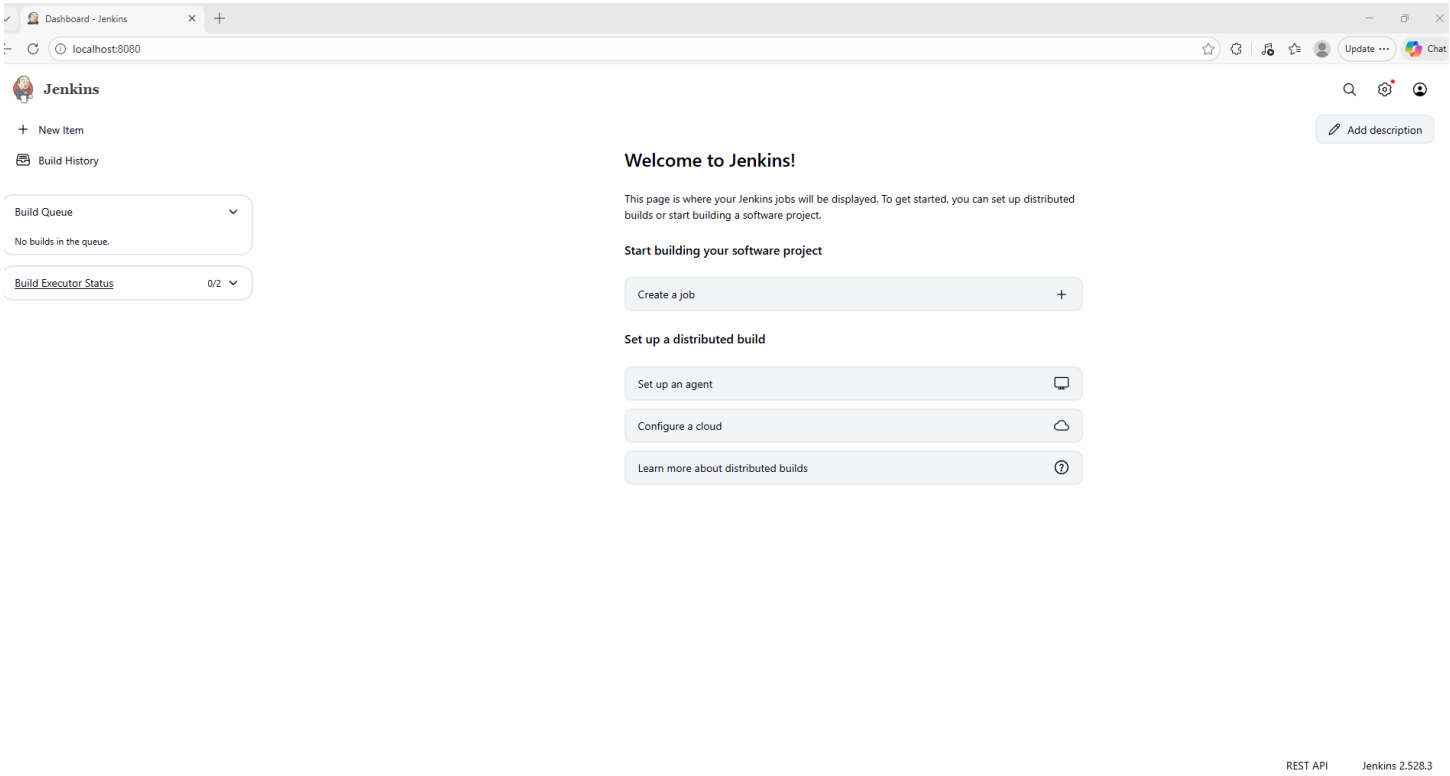
Configure a cloud

Learn more about distributed builds

[Dashboard - Jenkins]

[Jenkins for Beginners]

# Jenkins UI First Screen after logging in:



# Goto Settings and then Systems

Jenkins / Manage Jenkins / System

## System

Home directory ?

By default, Jenkins stores all of its data in this directory on the file system

```
C:\Users\sweta\.jenkins
```

System Message

This message will be displayed at the top of the Jenkins main page. This can be useful for posting notifications to your users

```
<b>This Jenkins server is for learning</b>
<h1>Hello Sweta</h>
```

# of executors

5

**This Jenkins server is for learning**

# Hello Sweta

Build Queue

Build Executor Status

(0 of 5 executors busy)

## Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

### Start building your software project

Create a job +

### Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

Jenkins / Manage Jenkins / Tools

# Tools

## Maven Configuration

Default settings provider

Use default maven settings

Default global settings provider

Use default maven global settings

## JDK installations

JDK installations

Edited

+ Add JDK

**JDK**

Name

JAVA_HOME

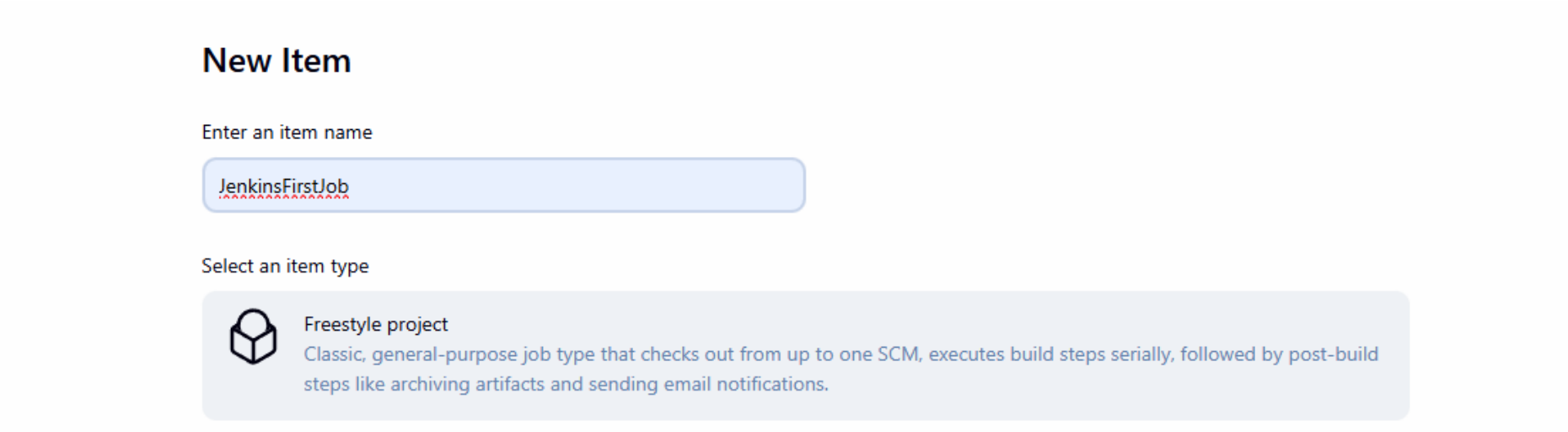
JAVA_HOME

C:\Program Files\Java\jdk-21

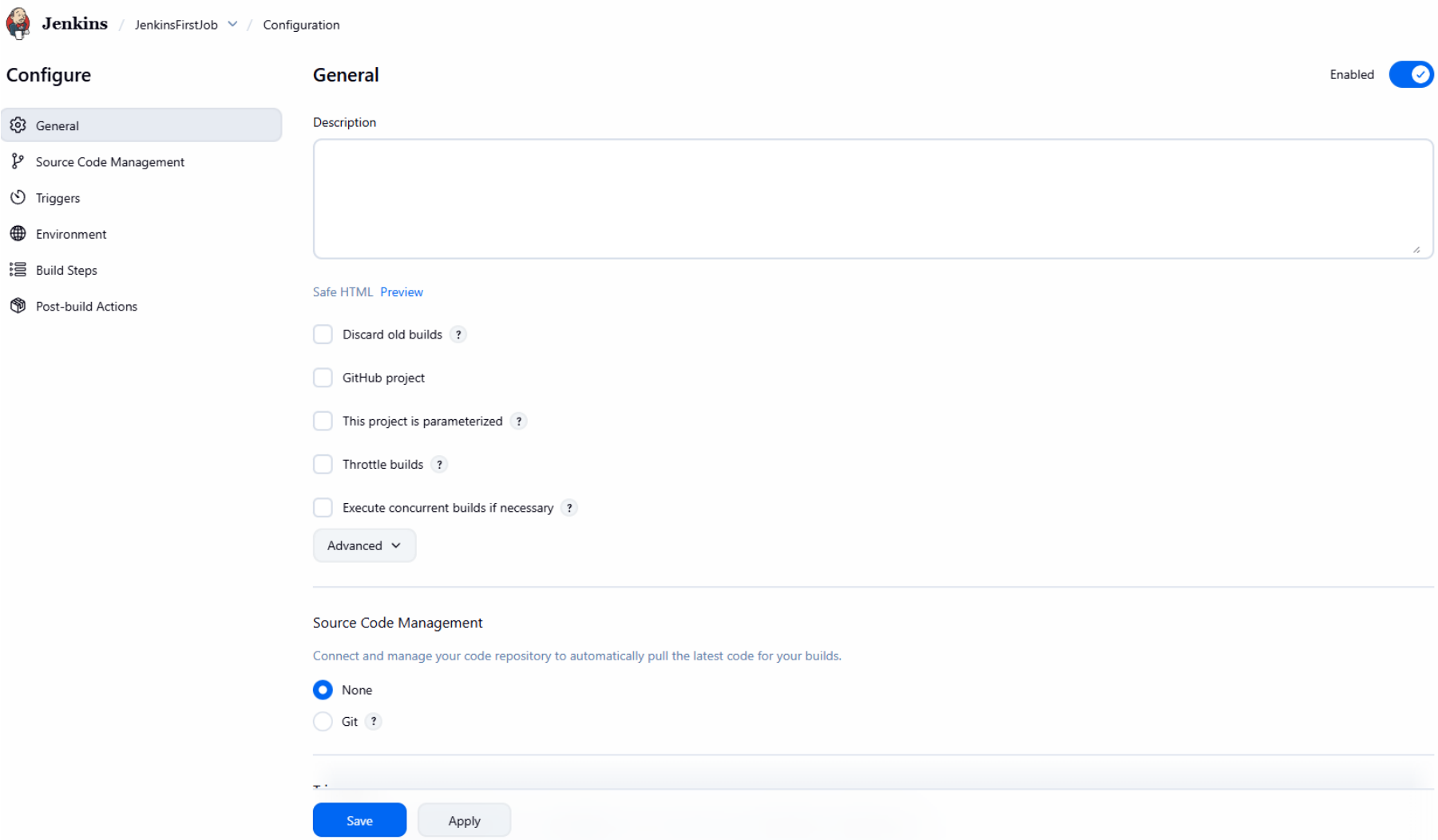
Install automatically ?

CREATING A JOB:

1. GO TO CREATE A JOB
2. AS AN EXAMPLE YOU MAY SELECT FREESTYLE PROJECT




3. CLICK OK

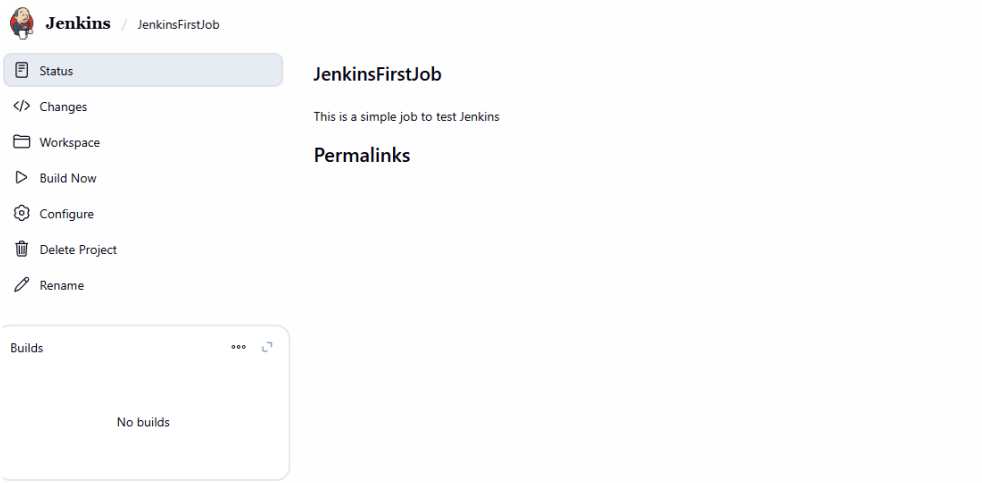



Click on Build Steps:

Select Execute Windows Batch Command

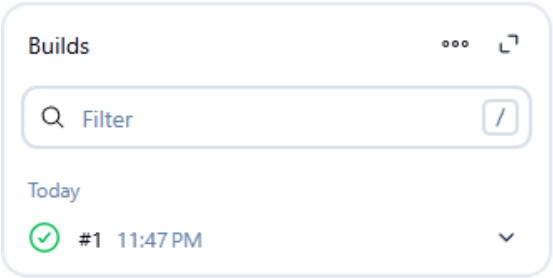
Inside the box -  you can write java -version

Apply




Click on Build now

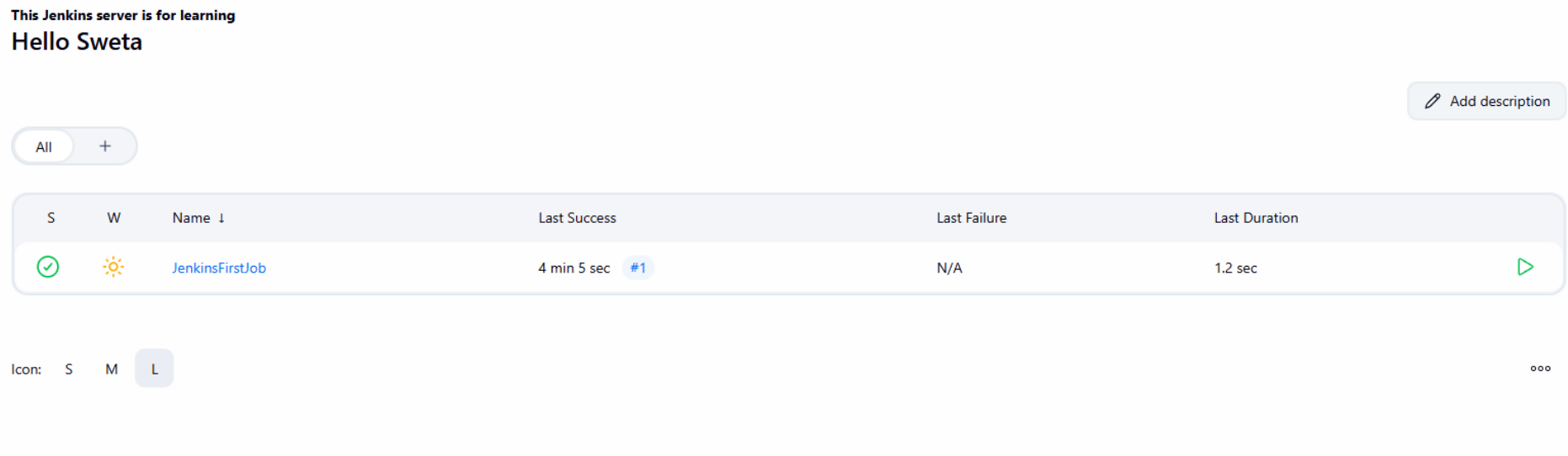
Click on the green tick:




You can view the console output:

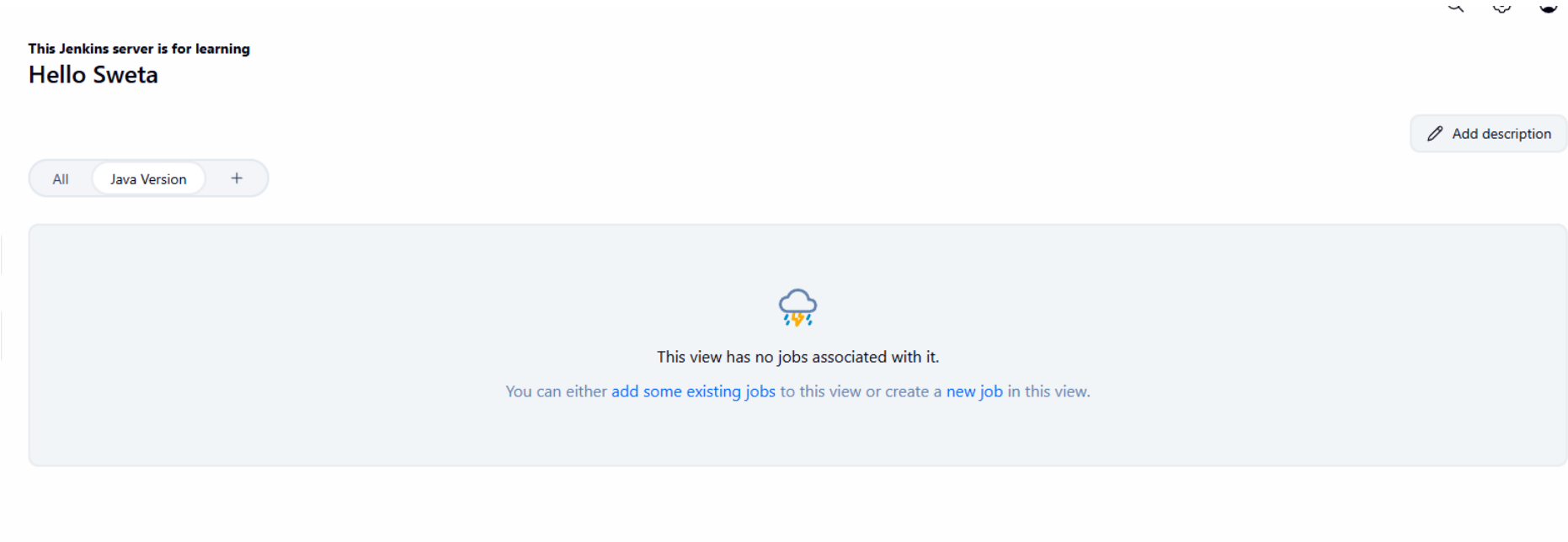
You can go to Dashboard by clicking on Jenkins on Left side

It will display the first job in the dashboard




You can change the view

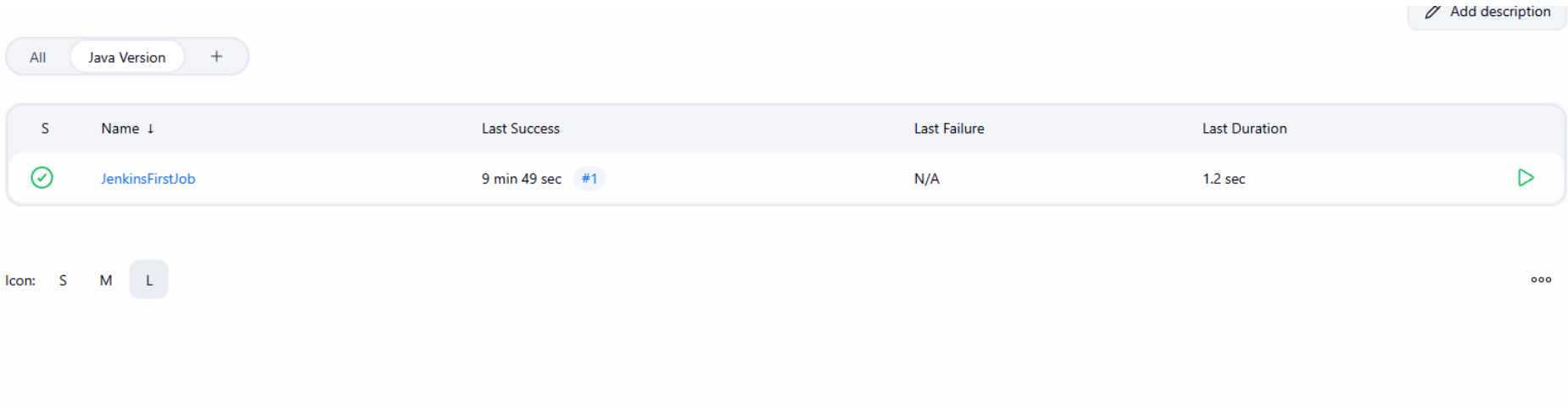
1. Click on Add View
2. Give a name of the view – Eg: Java Version
3. You can remove weather by clicking on the “x”
4. Apply and Save



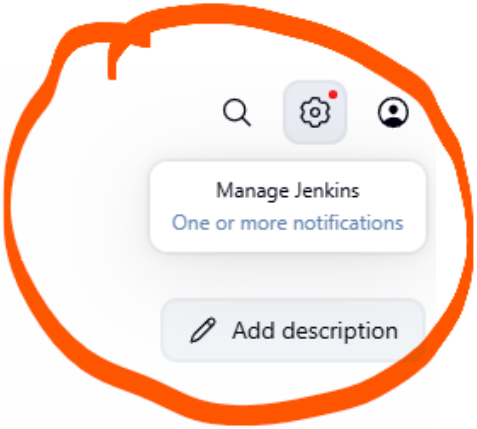

Click on Add some existing jobs:

Select the JenkinsFirstJob

Apply and save – The view without weather will appear as shown below:



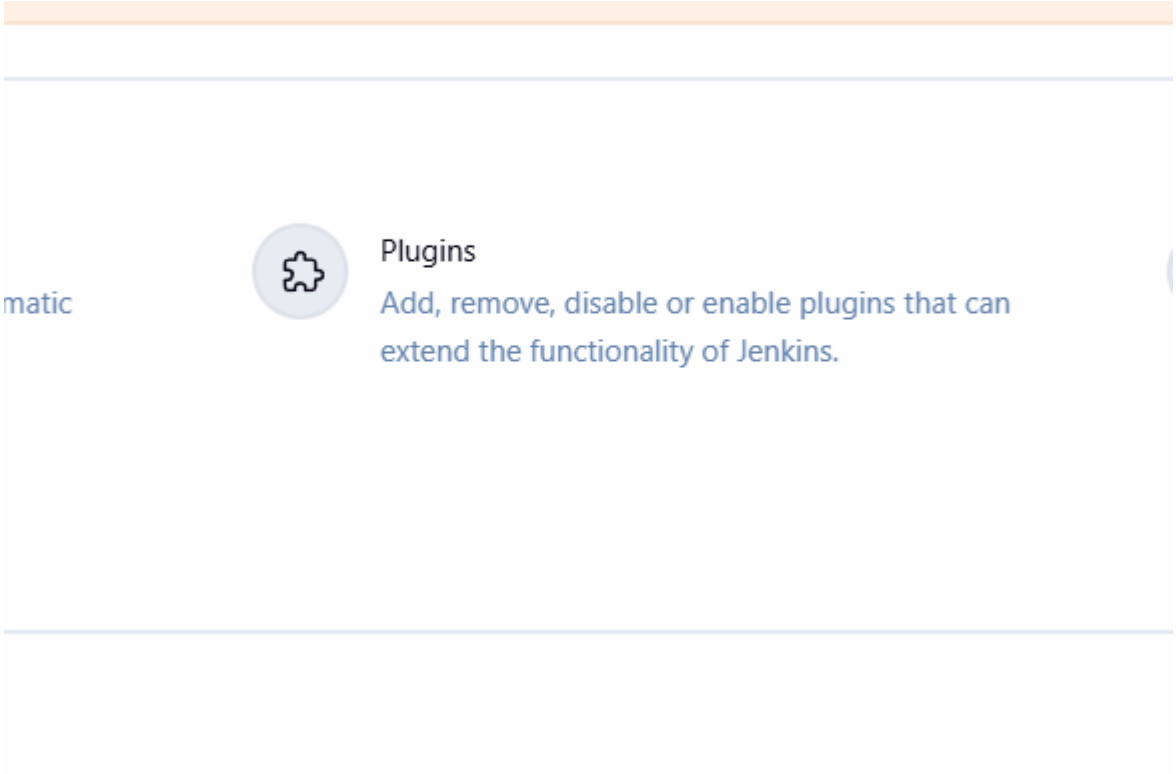

# Add Pluggins:




| Last Success | Last Failure | Last Duration |
| --- | --- | --- |
| 9 min 49 sec #1 | N/A | 1.2 sec |

...

Click on Plugins:



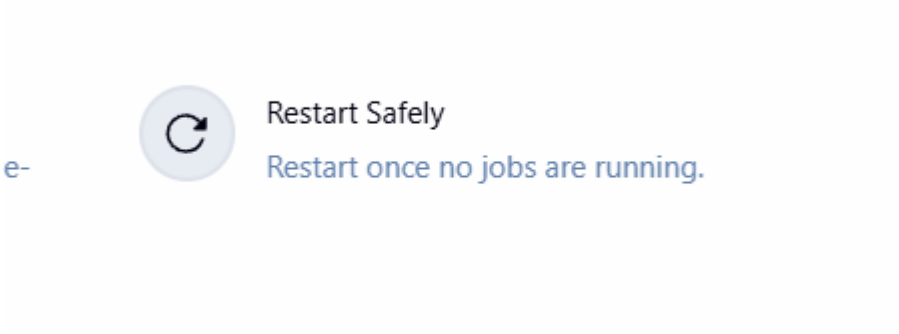

You can check installed plugins

You can click on available plugins – Type Safe restart

Select the check box on left of “Safe Restart” and select Install on Right

The plugin gets installed

The icon will appear in the Manage Jenkins




Now you can also Uninstall

Then restart Jenkins from the CMD prompt

Refresh the page

Then it will reappear under the Available Plugins

Jenkins / Manage Jenkins / Plugins

# Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings

safe /

| Install | Name ↓ | Released |
|---|---|---|
| ☐ | Safe Restart 102.v4dc1b_9636a_ee<br>Miscellaneous<br>This plugin allows you to restart Jenkins safely.<br>This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information. | 11 mo ago |

## Advanced settings

The Proxy configuration form has been moved to Configure System page

### Deploy Plugin

You can select a plugin file from your local system or provide a URL to install a plugin from outside the configured update site(s).

File

Choose File  No file chosen

Or

URL

Deploy

### Update Site

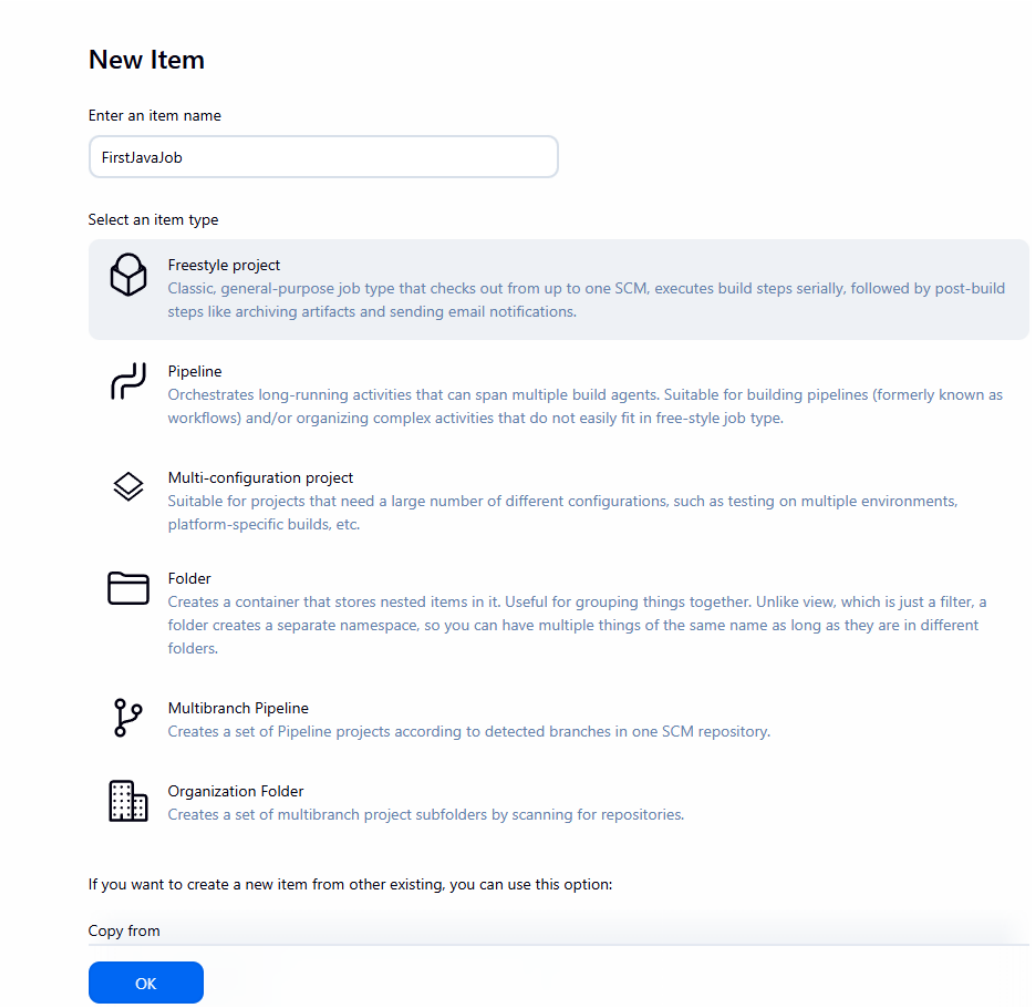
URL

https://updates.jenkins.io/update-center.json

Submit

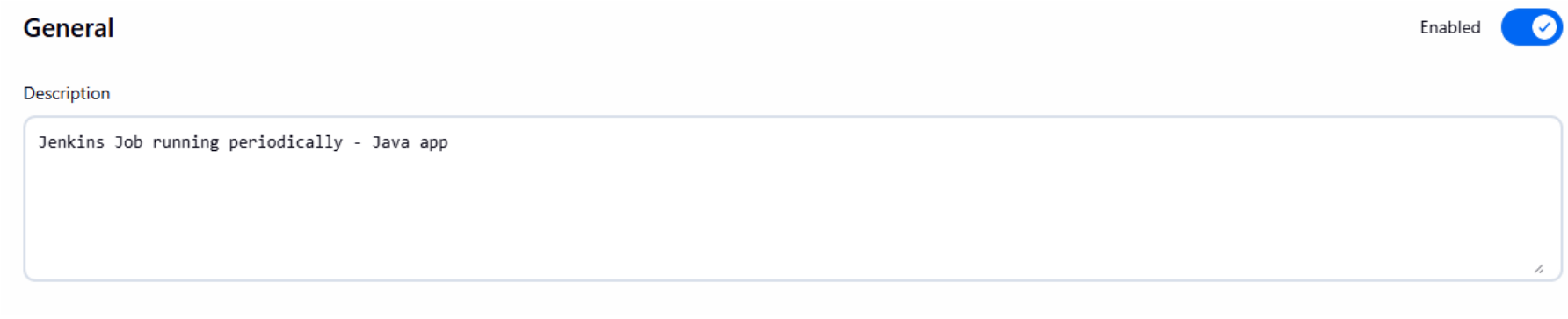
You can also install the plugin in your local machine and click on choose file

# Create a Java Program Job:

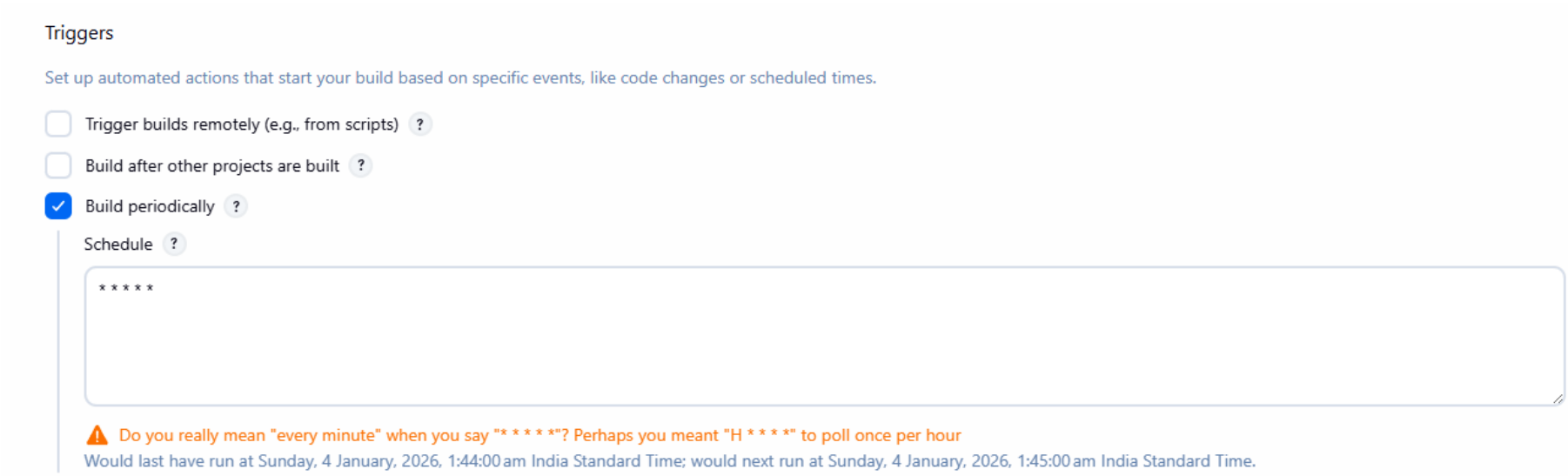



# Click OK

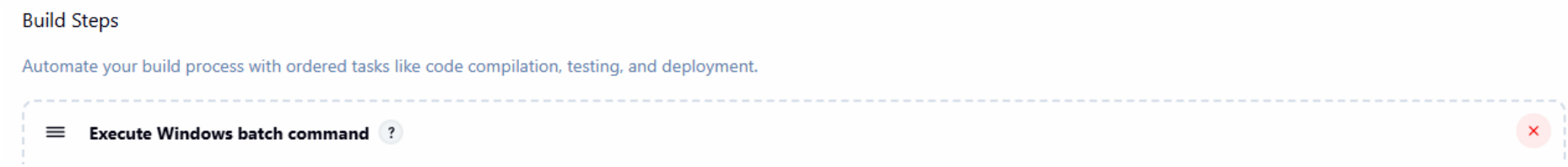
# Step 1: Add Description

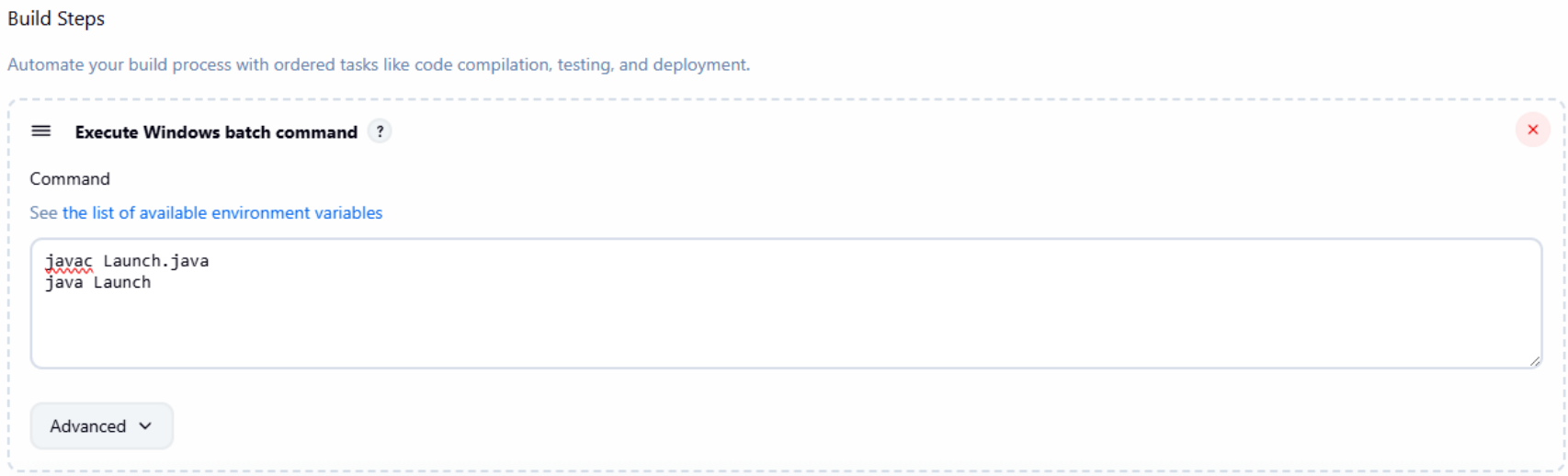



# Add Trigger:



# Add Build Steps – Windows Batch command

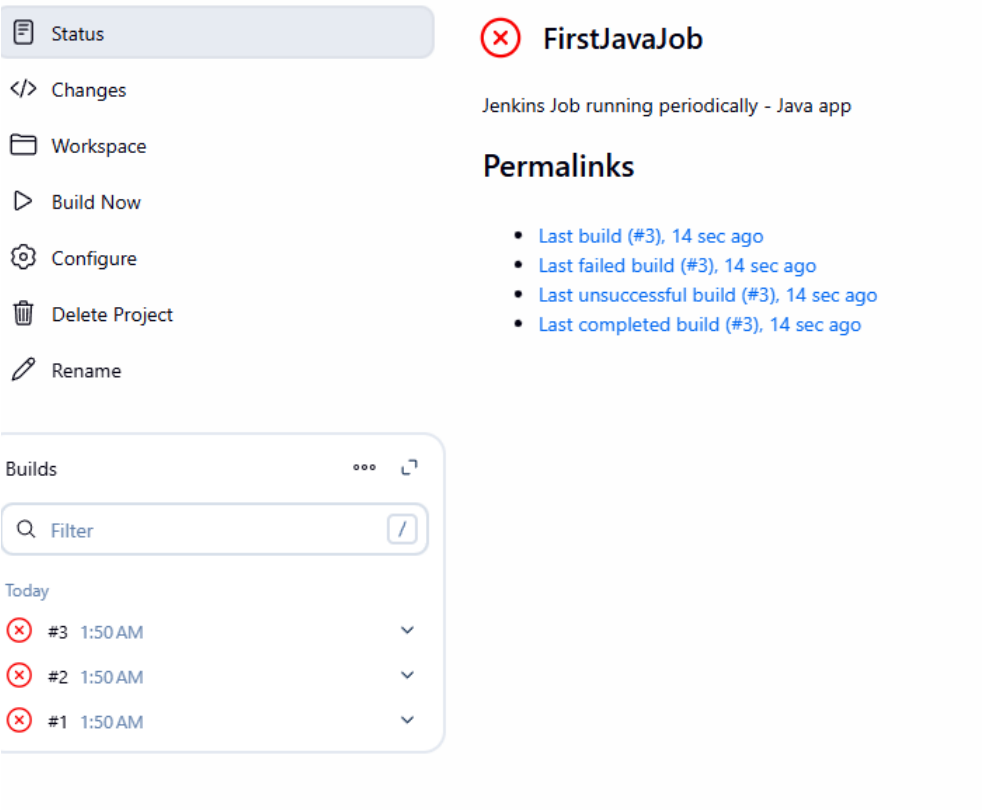



Click Apply ok

Come to – FirstJavaJob screen and click on Build Now

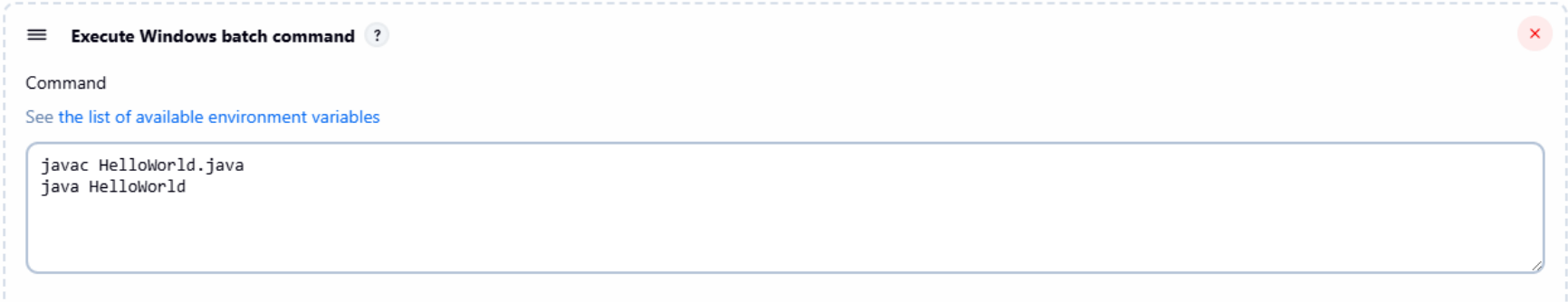
So the Build runs every second



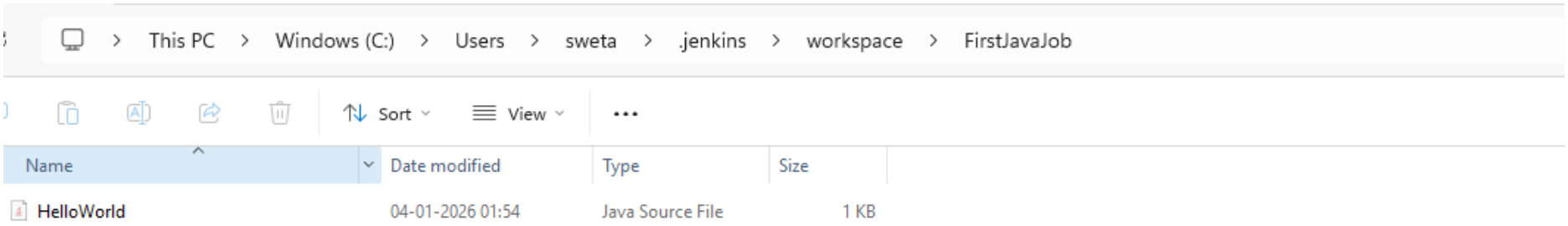

# Reason:

```
C:\Users\sweta\.jenkins\workspace\FirstJavaJob>java Launch
Error: Could not find or load main class Launch
Caused by: java.lang.ClassNotFoundException: Launch
```

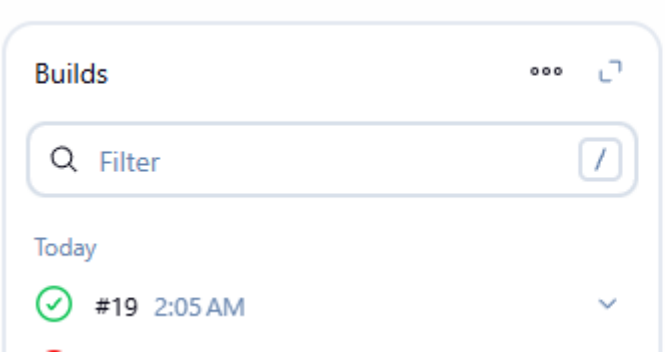
Now I create a helloworld program

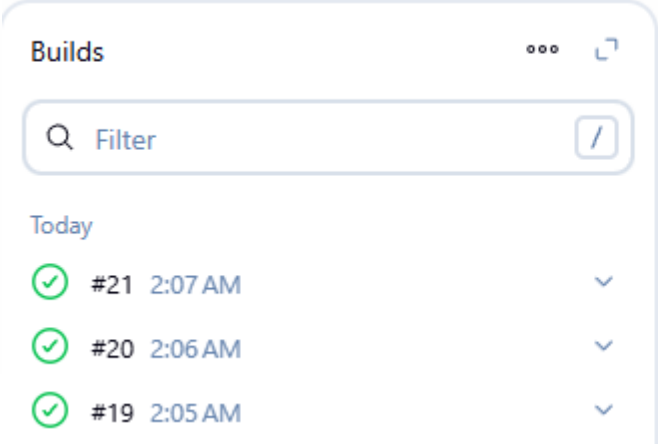



And then paste the .java file in the location where jenkins searches for the file –




Click om Build Now




Console Output:

## Console Output

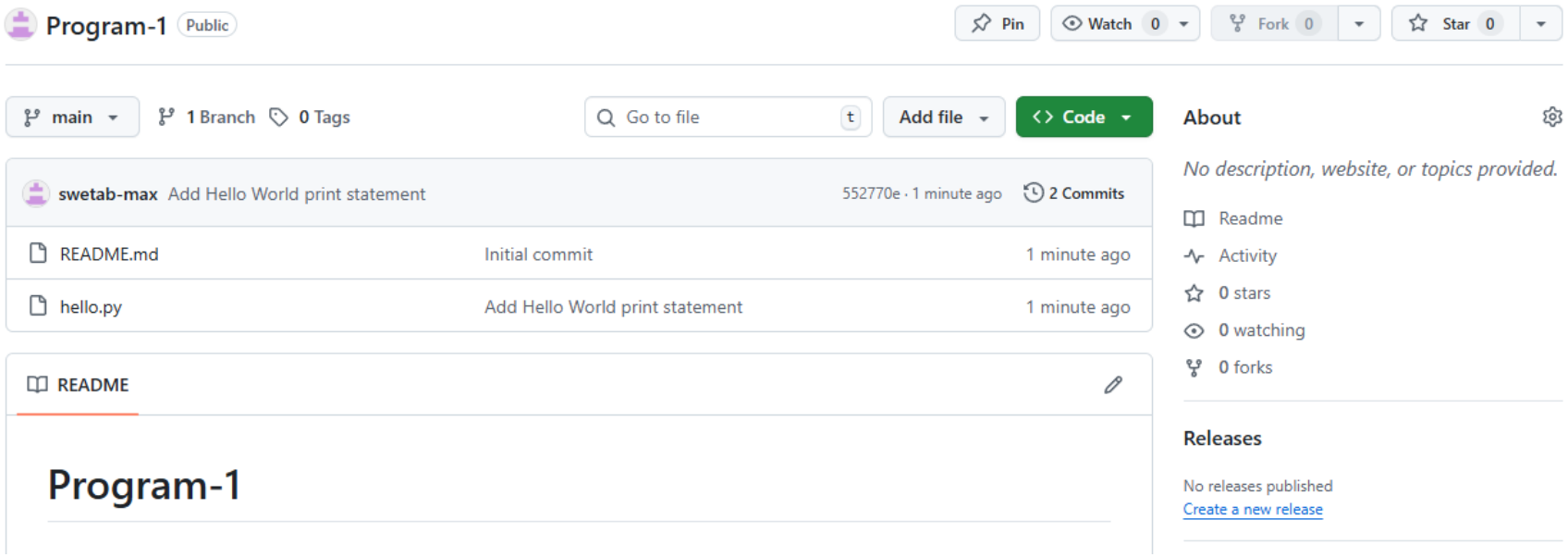
```
Started by timer
Running as SYSTEM
Building in workspace C:\Users\sweta\.jenkins\workspace\FirstJavaJob
[FirstJavaJob] $ cmd /c call C:\Users\sweta\AppData\Local\Temp\jenkins6552117785408533484.bat

C:\Users\sweta\.jenkins\workspace\FirstJavaJob>javac HelloWorld.java

C:\Users\sweta\.jenkins\workspace\FirstJavaJob>java HelloWorld
Hello, World!

C:\Users\sweta\.jenkins\workspace\FirstJavaJob>exit 0
Finished: SUCCESS
```

Create a Repository and add a program in GitHhub

We have created a Program-1 repository

We have further created a hello.py program inside it

Copy the link of the repository in Githib



Then in Jenkins:

Create New job as done earlier

Configure the job as shown below:

1. Add Description:

2. Select GitHub Project – Paste the GitHub Repo URL : https://github.com/swetab-max/Program-1.git/
3. Source Code Management – Paste the GitHub Repo URL : https://github.com/swetab-max/Program-1.git/

4. For Credentials – Click on Add and Select Jenkins
5. In the Jenkins Credential Provider Screen: Add Jenkins Username and Password

Jenkins Credentials Provider: Jenkins

## Add Credentials

Domain

Global credentials (unrestricted)

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

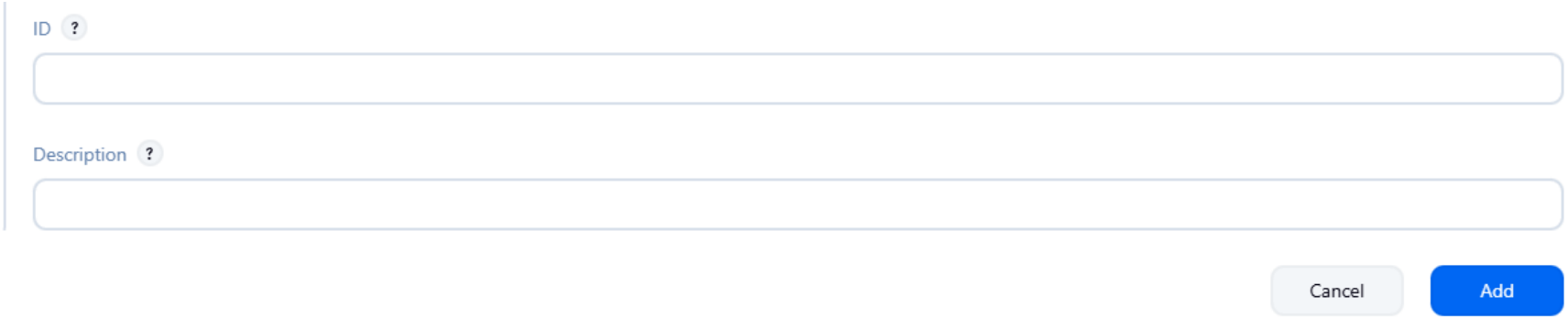
Username

Blank username; did you mean to use secret text credentials instead?

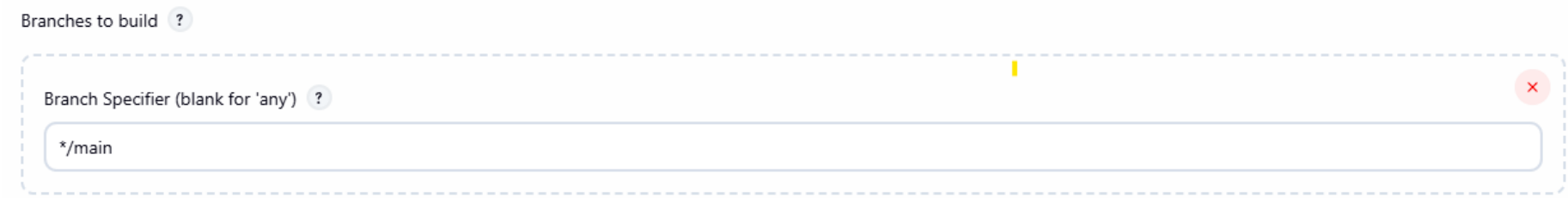
Treat username as secret ?

Password

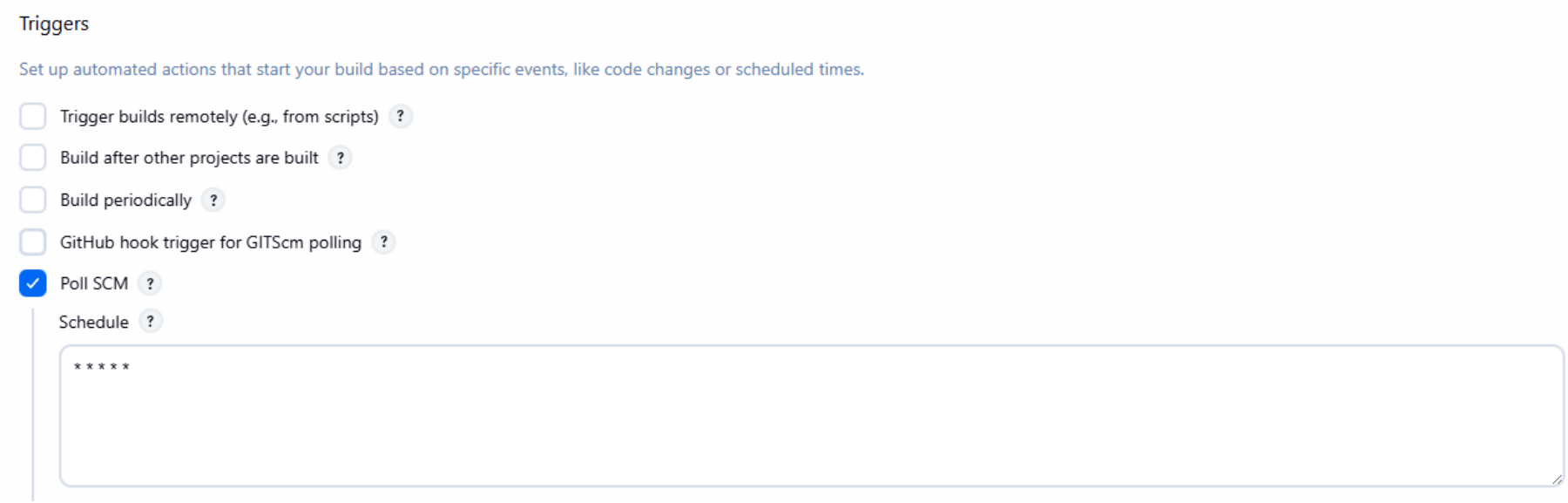
6. Give any ID and Description of your choice and click on Add




7. Select the Credential from the drop down. The credential you just created will be visible, just click on it and select
8. For Branches to Build - Since my program in under main so update to main removing master

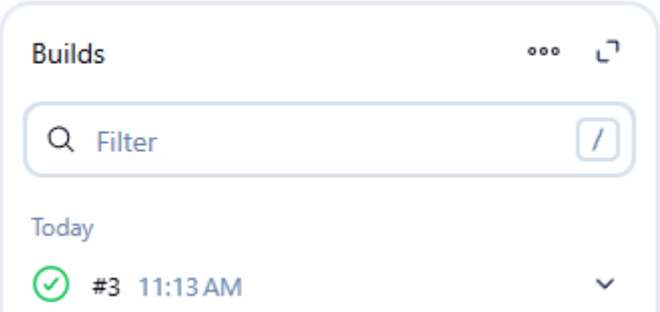


9. For Triggers – Select Poll SCM and specify the duration / frequency of running the build. In this case, any commit when done in Git will trigger the Build run here in Jenkins.




10. For Environment select Delete workspace before build starts
11. For Build Steps – Select Execute Windows Batch Command – Write the execution command for your program in Git – python hello.py
12. Apply and Save
13. Click on Build Now

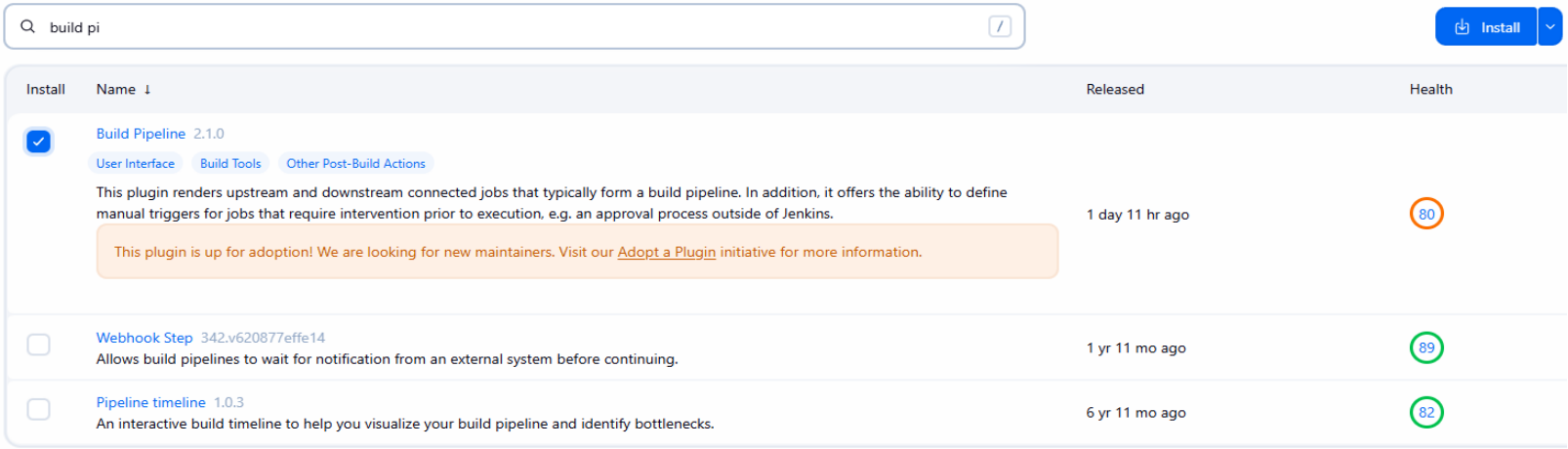
14. Build will run successfully and console Output will show Success

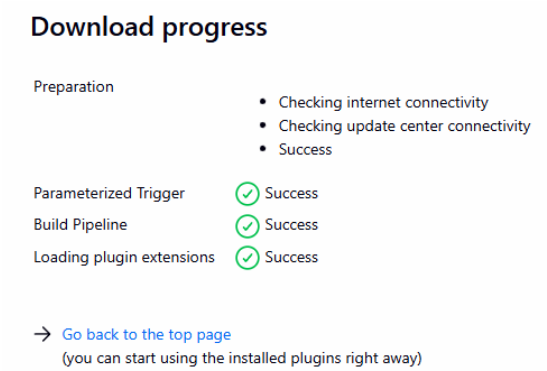



```
C:\Users\sweta\.jenkins\workspace\GITProject1>python hello.py
Hello, World!

C:\Users\sweta\.jenkins\workspace\GITProject1>exit 0
Finished: SUCCESS
```

# Build CI/CD Pipeline

1. Go to Manage Jenkins – Plugins – Available plugins
2. Search for Build Pipeline Plugin and install



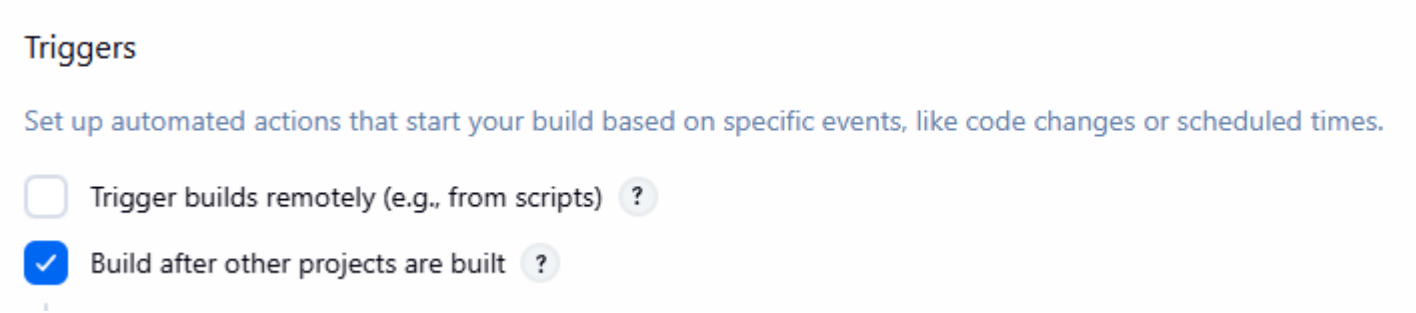



Then Create a New Job

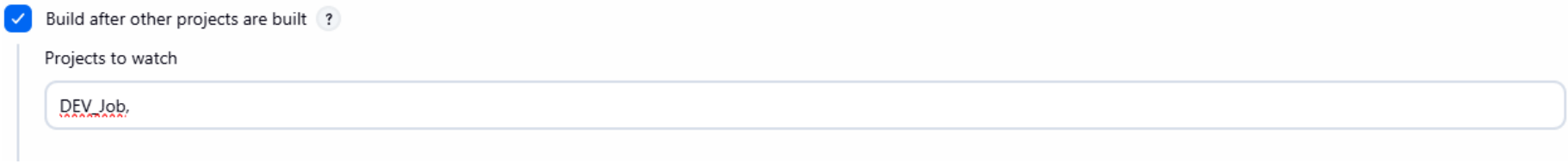
1. Create  Freestyle Job: DEV_Job
2. Follow the same steps in the previous section for adding the Git Repository URL and same URL in Source Code Management.
3. Select the same credentials that you selected earlier
4. Update the Branch Specifier to main
5. For Triggers – Select Poll SCM and add the five stars
6. For Environment – Delete workspace before build starts needs to be selected
7. For Build Steps select Execute Windows batch command
8. Type - python hello.py
9. Apply and Ok
10. The DEV_Job will be displayed in the dashboard

Then create another job – Name it: QA_Job, follow the same steps until Step 4.

5. For Triggers – Since this job should run only after DEV_Job runs successfully, like a pipeline, so select “Build after other projects are built”



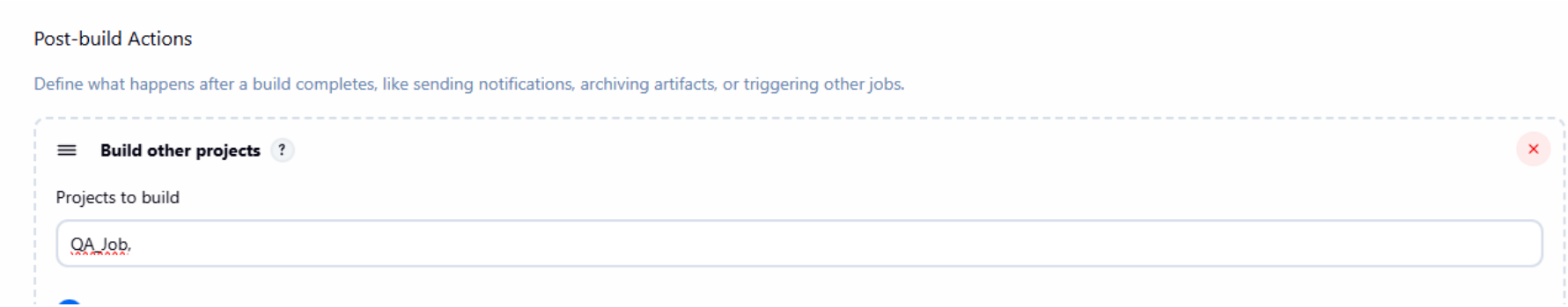

Select DEV_Job for Projects to watch



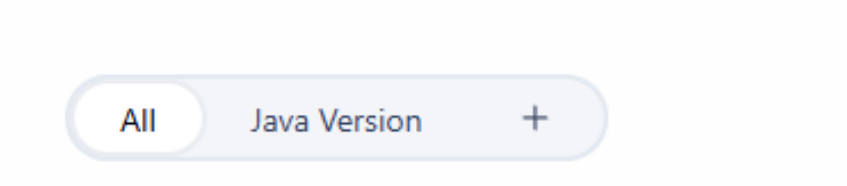

6. For Environment – Delete workspace before build starts needs to be selected
7. For Build Steps select Execute Windows batch command
8. Type - python add.py
9. Apply and Ok

10. The QA_Job will be displayed in the dashboard

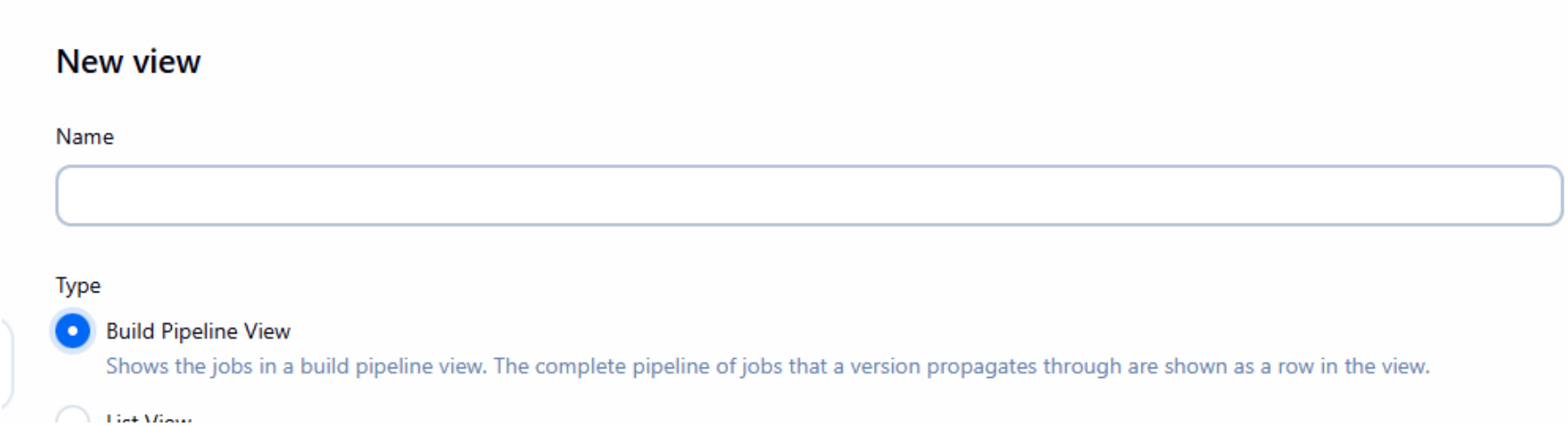
Now for the Dev_Job add:



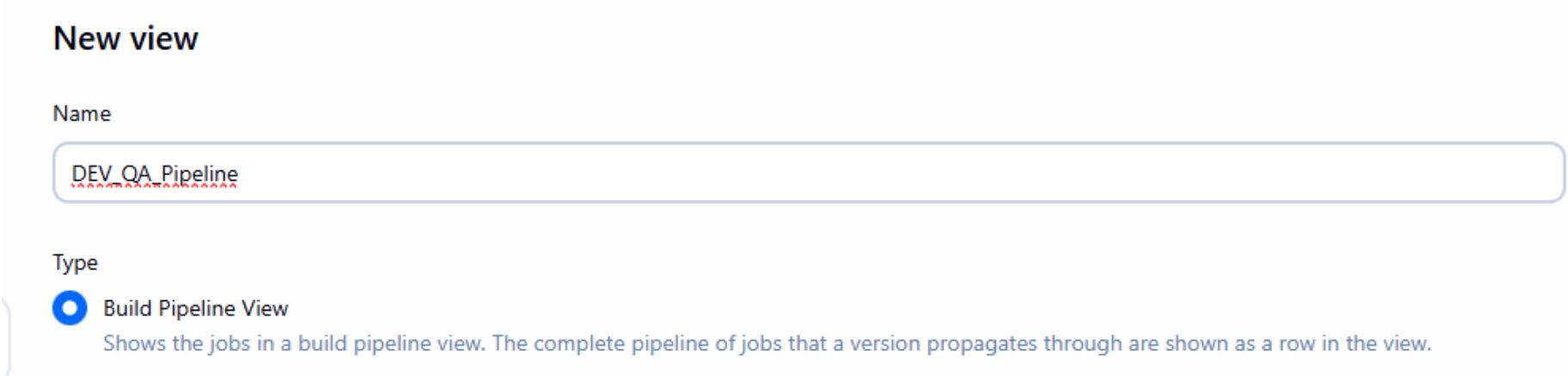

Now click on + sign to add a new View



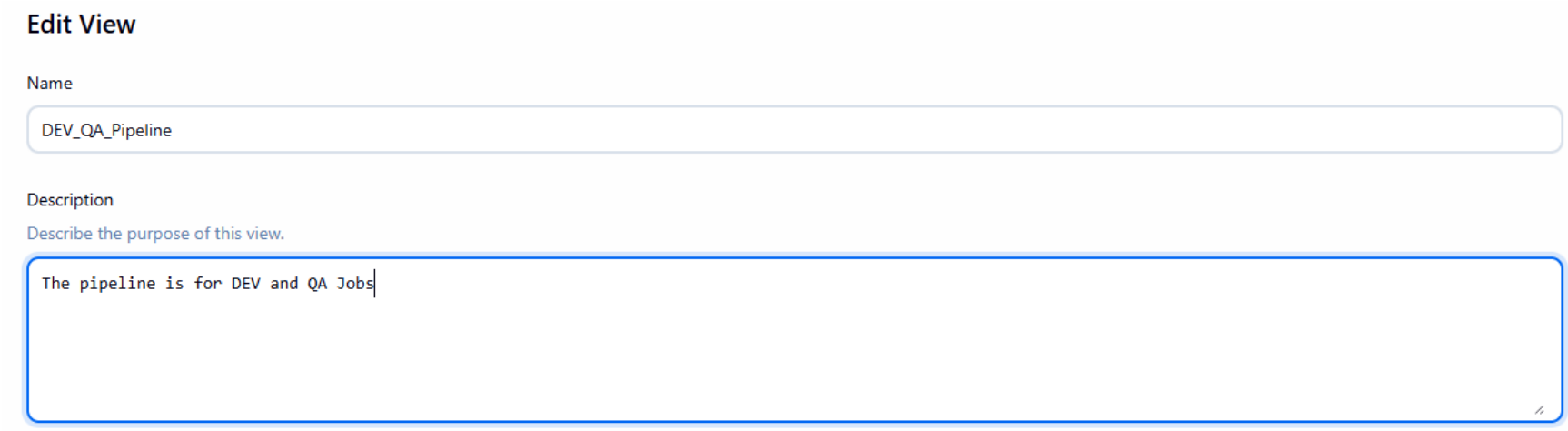

Select Build Pipeline View




Give the Name – I have given DEV_QA_Pipeline and click on Create



In the Edit View Window Add Description




In the pipeline flow: The first job or initial job should be selected which is our DEV_Job

Pipeline Flow

Layout

Based on upstream/downstream relationship

This layout mode derives the pipeline structure based on the upstream/downstream trigger relationship between jobs. This is the only out-of-the-box supported layout mode, but is open for extension.

Upstream / downstream config

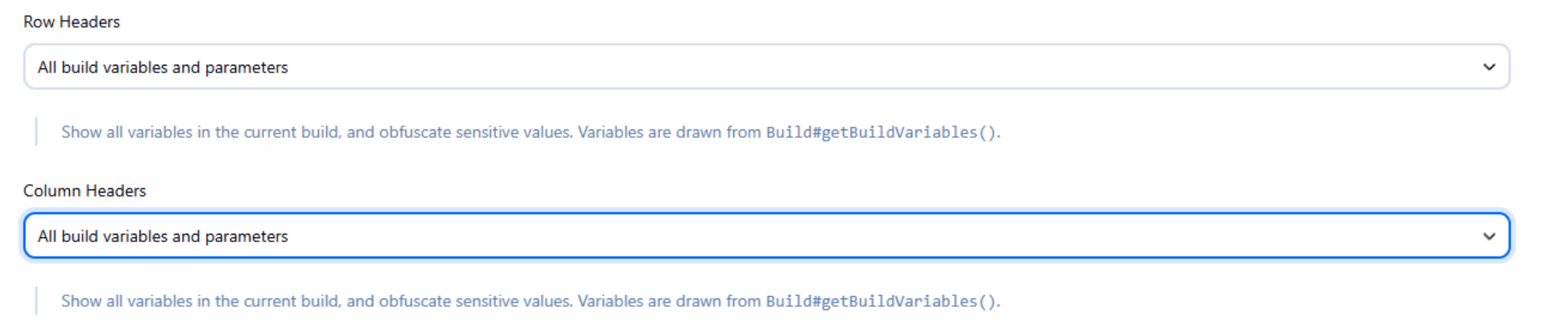
Select Initial Job ?

DEV_Job

No of Displayed Builds can be made as 5 or as per your requirement

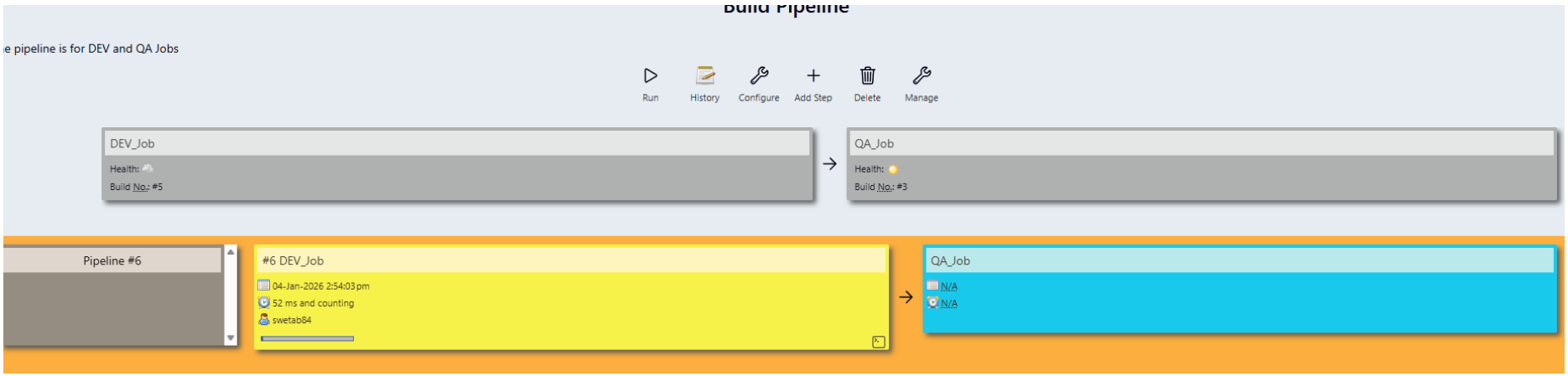
No Of Displayed Builds ?

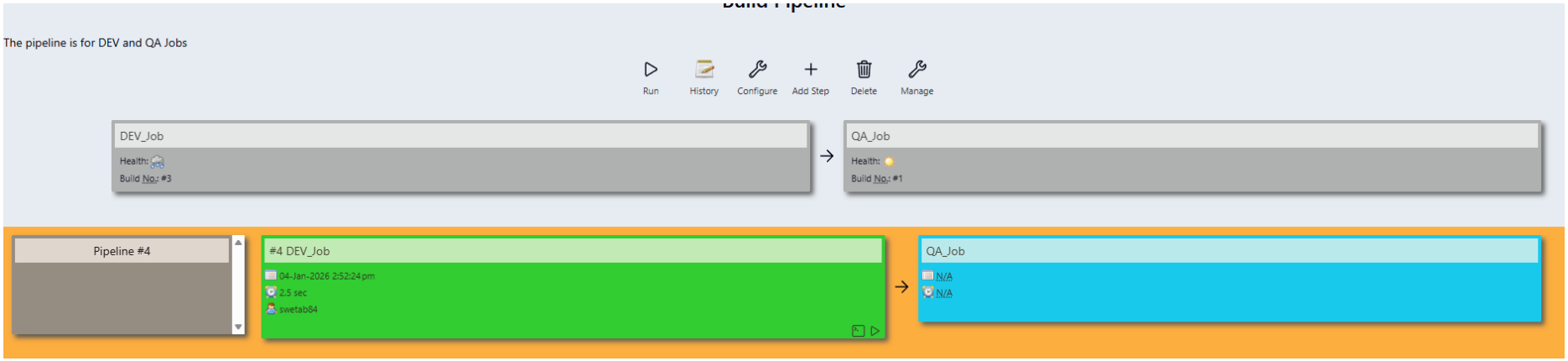
5

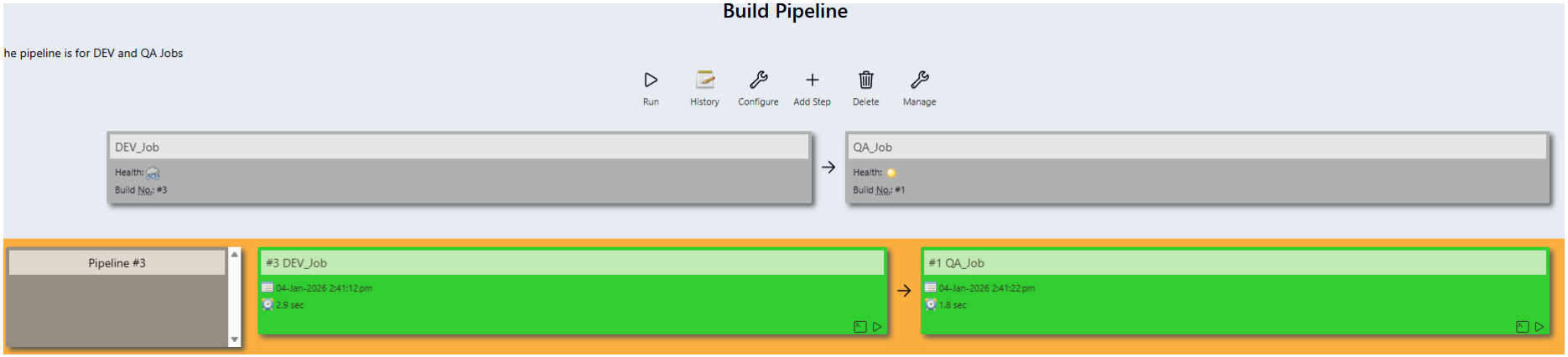



Click on Apply and Save

The Build Pipeline will be shown click on run






# Only when Dev_Job is executed then only QA_Job will run